In the Name of God

Deep Generative Models (25120)

Problem Set 3

Fall Semester 1404-05

Department of Electrical Engineering

Sharif University of Technology

Instructor: Dr. Sajjad Amini

Problem: 1×1 Convolution in Glow

In the Glow model, the authors introduced an invertible 1×1 convolution that generalizes channel permutations used in RealNVP. This operation improves flexibility and mixing across channels while maintaining an analytically tractable Jacobian determinant.

Let $X \in \mathbb{R}^{C\times H\times W}$ be the input tensor, and define the transformation:

$$Y_{i,j} = WX_{:,i,j}, \quad \text{where} \quad W \in \mathbb{R}^{C\times C},$$

applied independently to each spatial location (i, j).

(a) Jacobian and Log-Determinant

Find the Jacobian of this mapping, and prove that the transformation is invertible if and only if $\det\{W\} \neq 0$, and derive the explicit inverse:

$$X_{:,i,j} = W^{-1}Y_{i,j}.$$

(b) LU Parameterization

Glow parameterizes W as

$$W = P\,L\,(U + \text{diag}(\mathbf{s})),$$

where P is a fixed permutation matrix, L is lower triangular with ones on the diagonal, U is strictly upper triangular, and $\mathbf{s} \in \mathbb{R}^C$ contains diagonal entries.

(i) Prove that:

$$\log|\det\{W\}| = \sum_{i=1}^{C} \log|s_i|.$$

(c) Efficient Inverse

Show how to compute W^{-1} efficiently using the LU factors (without explicitly inverting W). Outline the algorithmic steps for solving $Wz = b$ using forward and backward substitution, and give the per-pixel computational complexity in terms of C.

Problem: Theory of Continuous Normalizing Flows

Let $z(0) \sim p(z(0))$ be the initial variable, and let $z(t)$ denote its state at time t under the ODE

$$\frac{dz}{dt} = f_\theta(z(t), t).$$

The goal is to show:

$$\frac{d}{dt} \log p(z(t)) = -\mathrm{Tr}\left(\frac{\partial f_\theta}{\partial z}\right).$$

We derive this step by step.

(a.i) Change of Variables Formula

Using the standard change-of-variables formula, write $\log p(z(t))$

(a.ii) Differentiate with Respect to Time

Differentiate the expression from part (a.i) with respect to t.

(a.iii) Apply Jacobi's Formula

You will obtain a term involving

$$\frac{d}{dt} \log |\det(J(t))|.$$

Jacobi's formula states:

$$\frac{d}{dt} \log |\det(A(t))| = \mathrm{Tr}\left(A(t)^{-1} \frac{dA(t)}{dt}\right).$$

Prove Jacobi's formula and apply it to express $\frac{d}{dt} \log p(z(t))$ in terms of a trace.

(a.iv) Find the Derivative of the Jacobian

To complete the expression you need $\frac{dJ(t)}{dt}$.

Prove that:

$$\frac{dJ(t)}{dt} = \frac{\partial f_\theta}{\partial z} J(t).$$

(a.v) Substitute and Conclude

Substitute your result for $\frac{dJ(t)}{dt}$ from (a.iv) into the trace expression from (a.iii).

Obtain the final formula:

$$\frac{d}{dt} \log p(z(t)) = -\mathrm{Tr}\left(\frac{\partial f_\theta}{\partial z}\right).$$

Problem: Duality in Autoregressive Normalizing Flows

Let $\pi_{\mathbf{x}}(\mathbf{x})$ be the true (unknown) data density over $\mathbb{R}^D$, and let $\pi_{\mathbf{u}}(\mathbf{u}) = \mathcal{N}(\mathbf{u} \mid \mathbf{0}, \mathbf{I})$ be the standard Gaussian base density. Consider a *stacked autoregressive normalizing flow* composed of K autoregressive layers, each parameterized via masked neural networks (MADE-style) with Gaussian conditionals.

(a) Theory of Masked Autoregressive Flow (MAF)

Derive the *forward transformation*, *inverse transformation*, and the *exact log-density expression* for a single MAF layer. Then, for a stacked MAF with K layers, formally define the full transformation $\mathbf{x} = f(\mathbf{u})$ and the log-density $\log p_{\text{MAF}}(\mathbf{x})$.

Explain why MAF enables *fast density evaluation* in a single forward pass, and why *sampling* requires D sequential autoregressive evaluations.

(b) Theory of Inverse Autoregressive Flow (IAF)

Derive the *forward transformation* and *inverse transformation* for a single IAF layer, and show how the conditioning is reversed compared to MAF.

Explain why IAF enables *fast sampling* in a single forward pass, but density evaluation of an external data point requires D sequential passes.

(c) Equivalence Between MAF and IAF

Let $f : \mathbf{u} \mapsto \mathbf{x}$ be the transformation implemented by a trained MAF model (fitted via maximum likelihood). Define the *implicit density* $p_{\mathbf{u}}(\mathbf{u})$ induced over the latent space by this trained transformation.

Prove that

$$\mathbb{D}_{\text{KL}}(\pi_{\mathbf{x}}(\mathbf{x}) \parallel p_{\text{MAF}}(\mathbf{x})) = \mathbb{D}_{\text{KL}}(p_{\mathbf{u}}(\mathbf{u}) \parallel \pi_{\mathbf{u}}(\mathbf{u}))\,.$$

Interpret this identity: training a MAF via maximum likelihood on $\pi_{\mathbf{x}}$ is equivalent to performing stochastic variational inference with an *implicit IAF* that uses $\pi_{\mathbf{x}}$ as its base density and f^{-1} as its reparameterization map.

Problem: VAEs with Conditional Flows

In a Variational Autoencoder, if we want to model the posterior as a more complicated distribution than a simple Gaussian, we can intuitively use a normalizing flow to transform a base Gaussian for better density approximation.

Let $\tilde{p}(x)$ be the empirical data distribution over $x \in \mathbb{R}^d$. We fit a generative model

$$q(x) = \int q(z)\, q(x|z)\, dz$$

with prior $q(z) = \mathcal{N}(0, I)$.

(a) Deriving the General Loss

Training minimizes the joint KL divergence

$$\mathcal{L} = \iint \tilde{p}(x)\, p(z|x)\, \log \frac{\tilde{p}(x)\, p(z|x)}{q(x|z)\, q(z)}\, dz\, dx. \tag{1}$$

Instead of a Gaussian posterior, define a **conditional flow posterior**

$$p(z|x) = \int \delta\big(z - F_x(u)\big)\, \mathcal{N}(u; 0, I)\, du, \tag{2}$$

where $F_x : \mathbb{R}^d \to \mathbb{R}^d$ is invertible in u for fixed x, with inverse $H_x = F_x^{-1}$.

(ai)

Substitute (2) into (1) and use the sifting property of the Dirac delta to eliminate the inner z-integral. Show that

$$\mathcal{L} = \iint \tilde{p}(x)\,\mathcal{N}(u;0,I)\,\log\frac{\tilde{p}(x)\,\mathcal{N}(u;0,I)}{q\big(x \mid F_x(u)\big)\,q\big(F_x(u)\big)}\left|\det\left\{\left(\frac{\partial F_x(u')}{\partial u'}\right)\right\}\right|^{-1}_{u'=H_x(F_x(u))} du\,dx. \quad (3)$$

(aii)

Perform the change of variables $v = F_x(u')$. Then $u' = H_x(v)$ and

$$\left|\det\left\{\left(\frac{\partial u'}{\partial v}\right)\right\}\right| = \frac{1}{\left|\det\left\{\left(\frac{\partial F_x(u')}{\partial u'}\right)\right\}\right|}. \quad (4)$$

Substitute, set $v = F_x(u)$ inside the delta, and simplify step-by-step to obtain

$$\mathcal{L} = \iint \tilde{p}(x)\,\mathcal{N}(u;0,I)\,\log\Big(\frac{\tilde{p}(x)\,\mathcal{N}(u;0,I)}{q\big(x \mid F_x(u)\big)\,\mathcal{N}\big(F_x(u);0,I\big)}\left|\det\left\{\left(\frac{\partial F_x(u)}{\partial u}\right)\right\}\right|^{-1}\Big)\,du\,dx. \quad (5)$$

(aiii)

Explain (i) why the Jacobian determinant appears in **the denominator**, and (ii) how the conditional flow $F_x(\cdot)$ makes $p(z \mid x)$ strictly more expressive than any diagonal Gaussian.

(b) Recovering Flow-Based Models

Consider the choice

$$F_x(u) = F(\sigma u + x), \qquad q(x \mid z) = \mathcal{N}\big(x; F^{-1}(z),\, \sigma I\big) \quad (6)$$

where F is an *unconditional invertible flow* (e.g., RealNVP/Glow) and $\sigma > 0$ is a small constant.

(bi)

Compute $\log q(x \mid F_x(u))$. Use invertibility of F to simplify and show

$$\log q(x \mid F_x(u)) = -\frac{d}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|u\|^2. \quad (7)$$

(bii)

Compute the Jacobian term using the chain rule. Let $w = \sigma u + x$, so $\frac{\partial w}{\partial u} = \sigma I$. Prove

$$\log\left|\det\left\{\left(\frac{\partial F_x(u)}{\partial u}\right)\right\}\right| = d\log\sigma + \log\left|\det\left\{\left(\frac{\partial F(w)}{\partial w}\right)\right\}\right|. \quad (8)$$

(biii)

Insert (7) and (5). Demonstrate that the reconstruction term $\log q(x \mid F_x(u))$ contains **no trainable parameters** in F or x. The loss then collapses (up to a constant) to

$$\mathcal{L} = \mathbb{E}_{\tilde{p}(x),\,u\sim\mathcal{N}(0,I)}\left[\frac{1}{2}\left\|F(\sigma u + x)\right\|^2 \;-\; \log\left|\det\left\{\left(\frac{\partial F(\sigma u + x)}{\partial u}\right)\right\}\right|\right] \;+\; \text{const.} \quad (9)$$

Conclude that f-VAE recovers **standard flow-based training with input dequantization noise** σ.

$*$ VP-NFs are not universal

Remember a Normalizing Flow formulated as:

$$p_\theta(\mathbf{x}) = p\big(z = f_\theta^{-1}(\mathbf{x})\big) \left|\det\left\{\frac{\partial f_\theta^{-1}(\mathbf{x})}{\partial \mathbf{x}}\right\}\right| \tag{1}$$

In a Volume-Preserving Normalizing Flow (VP-NF), the determinant of the Jacobian matrix of each transformation is constant, specifically:

$$\left|\det\left\{\frac{\partial f_\theta(\mathbf{z})}{\partial \mathbf{z}}\right\}\right| = 1$$

This implies that the transformation preserves volume in the latent space, meaning the model can rearrange the probability mass but cannot scale it.

Definition: A set of probability distributions $\mathcal{P}$ is called a distributional universal approximator if for every possible target distribution $p(x)$ there is a sequence of distributions $p_n(x) \in \mathcal{P}$ such that $\lim_{n\to\infty} p_n(x) \sim p(x)$ with a chosen type of convergence.

We want to prove: The family of Normalizing Flows with constant Jacobian determinant $\left|\det\left\{\frac{\partial f_\theta(\mathbf{z})}{\partial \mathbf{z}}\right\}\right| = \text{const}$, with latent $z \sim \mathcal{N}(0,1)$ is not a universal distribution approximator under KL divergence. We do this by finding a lower bound for the KL objective. Use the Pinsker's inequality as below and the counter example given:

$$\delta(p_\theta, q) = \sup_{E\text{: a measurable event}} \{|p_\theta(E) - q(E)|\} \leq \sqrt{\frac{1}{2} D_{\mathrm{KL}}(p\|q)}$$

counter example:

$$p(x,y) = \begin{cases} 0.9 & \text{if } (x,y) \in [-0.5, 0.5] \times [-0.5, 0.5], \\ 0.9 - k\cdot(|x| - 0.5) & \text{if } |x| \in [0.5, 0.9k + 0.5] \text{ and } |y| \in [0, |x|], \\ 0.9 - k\cdot(|y| - 0.5) & \text{if } |y| \in [0.5, 0.9k + 0.5] \text{ and } |x| \in [0, |y|], \\ 0 & \text{otherwise.} \end{cases}$$

$$A = \left\{(x,y) \in \mathbb{R}^2 : p_\theta(x,y) \geq 0.9 - \epsilon\right\}$$
$$B = [-0.5, 0.5] \times [-0.5, 0.5]$$
$$\bar{A} = B \setminus A$$

Part (a)

Prove there exist an event E such that $|p_\theta(E) - p(E)| > 0$ when $A = \emptyset$, and show the lower bound for KL divergence can be positive for some $\epsilon > 0$, using Pinkster's inequality. (hint: prove $\bar{A}$ can be such E)

Part(b)

Using the change of variables formula, create the event C in latent space resulting from A. Compare its volume with A. Then compute its volume and provide a simple upper bound for it. (hint: volume here is a 2D circle because of the normal distribution)

Part(c)

Now show that there is an event E such that $|p_\theta(E) - p(E)| > 0$ when $A \neq \emptyset$, and show the lower bound for KL divergence can be positive for some $\epsilon > 0$, using Pinkster's inequality

Part(d)

Now briefly explain why this means VP-NFs cant approximate every distribution.